ODCE Arduino Scenario Kit

Guide for the Arduino Scenarios A, B, and C

The scenarios are designed but have not been run with students yet. The kit is supposed to be treated as a proposal. The development of this kit is based on the experience of a prior Arduino session run in the ODCE module in April 2026 and on the results of the master thesis "Planspiele als Vermittlungsinstrument organisationaler Problemlösungskompetenzen" by Christian Nguyen (TU Hamburg, 2026).

Implementation in ODCE

Each scenario is a small Arduino build that teams of **4 to 5 students** do twice. Between the two rounds one organizational parameter changes, while the task stays the same.

Scenario	Parameter	Round 1 to Round 2
A Conveyor Line	Interdependence	Pooled to sequential
B Greenhouse Control	Coordination Mechanism	Mutual adjustment to standardization
C Parking Garage	System Architecture	Integral to Modular

Implementation Options

Option	Pros	Cons
One scenario for all groups in one session (recommended)	- One set of materials - One focus for observation - Shared debriefing, because all groups compare the same two rounds.	Students only examine one parameter
All scenarios in one session w/ one scenario per group	Students hear about all parameters in the debriefing	Needs one tutor per scenario and debriefing gets harder because groups compare different things
One session per scenario	Students meet all three parameters themselves	Too much course time and contrast gets weaker because students know the pattern

Scenario Evaluation

- Scenario C is the best start. It is the closest to the previous session in April. Tt shows the shift from global to local fault search most clearly.
- Scenario A fits a lecture on interdependence. It has the simplest story. When tasks and stations are linked to each other, teams have to agree on handovers.
- Scenario B fits a lecture on coordination mechanisms and standards. Round 2 is a different task of the same kind. Parameter is not separated strictly as in A and C

Session Agenda

Phase	Minutes	Note
Briefing and groups	10	Hand out the Round 1 sheet only
Round 1	60	Teams work, observer examines
Team talk and break	15	Questions at the end of the sheet
Round 2 briefing	10	Hand out the Round 2 sheet
Round 2	60	Teams work, observer examines
Team talk	10	Questions at the end of the sheet
Debriefing	15 to 20	All groups together, instructor documents and takes notes
Feedback survey	5	Before students leave

Material

- Task sheets for both rounds, starter code, one kit per group
- Seeded_Faults.md (for instructor)
- Observation_Sheet.pdf, one per observer
- Feedback_Survey.pdf, one per student (transfer to a digital instrument such as Google Forms is recommended!)
- One observer for every 2 to 3 groups, and one note taker for the debriefing

Evaluation

The scenario kit examines five dimensions. D1 role clarity, D2 communication structure, D3 coordination quality, D4 fault diagnosis and D5 time and resources.

- **Observation**: Examine each group about every 10 minutes for 2 to 3 minutes. Note what is seen per dimension on the sheet. Note for each seeded fault who found it and whether the search was global or local.
- **Debriefing notes:** Show how students explain the difference between the rounds.
- **Survey:** Items F1 to F10 match the April 2026 survey, so results stay comparable.

Compare Round 1 and Round 2 within each group. Keep in mind that the order is fixed and students get better with practice. A change between rounds is therefore always a hint and not evidence!

Scenario A Conveyor Line

Description

Teams build three stations of a small conveyor line on one Arduino. A detection station notices a part, a sorting station sends it to lane A or B, and a counting station counts it. Parts, wiring, roles and time are the same in both rounds. The scenario shows that the need for coordination comes from how tasks depend on each other.

Changes between both Rounds

Round	Setup
Round 1	Pooled: Each station works on its own and needs nothing from the others
Round 2	Sequential: A station may only work on a part the station before it has handed over. The last 15 minutes are for connecting and testing the full line.

Structure

- Roles are **line coordinator**, **detection developer**, **sorting developer**, **counting developer** and **tester**. In a team of four the coordinator also tests
- Parts per group are an **Arduino Uno**, **one ultrasonic sensor**, **one light sensor**, **two buttons** and **four LEDs**
- Seeded fault A1: Sorting compares the wrong way, so bright parts go to lane B

Evaluation

- **D3 – Coordination:** Waiting at handovers and work done twice once the stations are linked
- **D2 – Communication:** Talk moves from inside a station to the handover and integration points
- **D4 – Fault Diagnosis:** In Round 2 a fault in sorting shows up later in counting. Note whether and where teams trace it back

Debriefing questions

- Where did you have to talk to another station in Round 2, and why?
- What happened when one station was late or wrong?
- Where in a real company setting does one team wait for the output of another?

Scenario B Greenhouse Control

Description

Teams build a small greenhouse control. In Round 1 it is a temperature control with a fan LED and a warning LED. In Round 2 it is a soil moisture control with a pump LED and two status LEDs. Both follow the same pattern of sensor, rule, actuator and report. Roles and time stay the same. The scenario shows what a standard gives a team and what it costs.

Changes between Rounds

Round	Setup
Round 1	Mutual adjustment. No standard is given. The team agrees on pins, thresholds and code as it goes.
Round 2	Standardization. A given standard fixes pins, value bands, message format and the order of steps. The team builds exactly to it.

Structure

- Roles are **coordinator**, **sensor developer**, **actuator developer**, **display and report developer**, and **tester**. In a team of four the coordinator also tests
- Round 1 needs a **TMP36 temperature sensor** and **two LEDs**. Round 2 needs a **potentiometer** and **three LEDs**. Hand out the Round 2 parts with the Round 2 sheet (avoid the time to look for the right parts)
- Seeded fault B1: An actuator follows the wrong rule. It is the same kind of fault in both rounds but sits in a different line of code

Evaluation

- **D2 – Communication:** Much talk about pins and thresholds in Round 1, less in Round 2
- **D3 – Coordination:** Parts that do not fit together in Round 1, and whether anyone checks the work against the standard in Round 2.
- **D1 – Role clarity:** Whether the tester role actually checks against the standard.

Debriefing questions

- What did the standard save you in Round 2, and what did it cost?
- Did anyone check the result against the standard?
- When does a real team need a standard, and when can it just agree as it goes?

Scenario C Parking Garage

Description

Teams build a guidance system for a small parking garage. Two ultrasonic sensors watch the entry and the exit. The system counts cars, refuses new cars when the garage is full and shows a traffic light. Function, parts, wiring, roles and time are the same in both rounds. The scenario shows how the structure of a system shapes teamwork and fault search.

Changes between Rounds

Round	Setup
Round 1	Integral: The team works on one shared program with no rules on who changes which part.
Round 2	Modular: The program is split into modules. The team first agrees on shared properties such as pins and functions and then leaves it unchanged. The last 15 minutes are for integration.

Evaluation

- **D4 – Fault diagnosis:** Global search through the whole program in Round 1, local search by one module owner in Round 2
- **D1 – Role clarity:** Whether owners stay in their module and leave the settings alone
- **D5 – Time:** Whether teams keep time for integration or squeeze it into the last minutes

Structure

- Roles are **team lead**, **detection developer**, **counting developer**, **display developer and integrator**. In a team of four the team lead also integrates
- Parts per group are an **Arduino Uno**, **two ultrasonic sensors** and **three LEDs (green, yellow, red)**. (Hint: Give each group a small toy car to act as a car)
- Seeded faults C1 to C3: A car is counted twice, the count drops below zero, and yellow and red light up together when the garage is full

Debriefing questions

- How did you find faults in Round 1, and how in Round 2?
- What did the shared settings cost you, and what did it give you?
- Where do real projects split work into modules, and what goes wrong at the interfaces?